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RESEARCH

Comparative effects of low-level laser acupuncture and dry needling on clinical and functional outcomes in patients with chronic cervical myofascial pain syndrome: a randomized controlled trial

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Abstract

The purpose of this study was to compare the effectiveness of Low-Level Laser Acupuncture with Dry Needling along with Routine Physical Therapy on clinical and functional outcomes in patients with Chronic Cervical Myofascial Pain Syndrome. An Assessor-blind Randomized Controlled trial was conducted at the Physical Therapy department of Rawal General & Dental Hospital, Pakistan. The study duration was June 2023 to November 2024. A non-probability convenience sampling technique was employed to produce a study sample of 100 individuals with Chronic Cervical Myofascial Pain Syndrome. The 100 eligible participants, with a mean age of 34 ± 7.21 years, were randomly allocated into two equal groups ($n = 50$ per group). Group A received Low-Level Laser Acupuncture and Routine Physical Therapy, while Group B underwent Dry Needling and Routine Physical Therapy. The outcomes of the study were pain intensity, neck disability, cervical range of motion, and health-related quality of life measured by the Numeric Pain Rating Scale, Neck Disability Index, Goniometer, and SF-36 health survey questionnaire, respectively. The outcome measures were evaluated at three temporal points: the pre, mid, and post-treatment sessions. Each participant received a total of 18 treatment sessions. The IBM SPSS Statistics version 26.0 was used to analyze and interpret the results. The Mann–Whitney U test, Friedman test, and subsequent Wilcoxon signed-rank post hoc analyses with Bonferroni adjustment demonstrated statistically significant differences ($p < 0.05$ in all cases), both intergroup and intragroup over time. The Low-Level Laser Acupuncture and Dry Needling both proved to be efficient adjunct therapies to routine physical therapy for Chronic Cervical Myofascial Pain Syndrome for different durations of improvement.

Keywords Chronic Cervical Myofascial Pain Syndrome (CMPS) · Low-Level Laser Acupuncture (LLLA) · Dry Needling (DN) · Cervical mobility · Photobiomodulation Therapy

Introduction

Chronic Cervical Myofascial Pain Syndrome (CMPS) is a multifactorial neuromuscular disorder characterized by the presence of hyperirritable myofascial trigger points (MTrPs) within taut bands of skeletal muscle fibers in the cervical region [1]. Myofascial trigger points are usually accompanied by localized tenderness, referral pain, reduced mobility, and functional disability, and can also be responsible for triggering other conditions like headaches, vertigo, and blurred vision. The pathogenesis of CMPS is believed to be multifaceted and is due to the synergistic effect of biomechanical loading, neuro-muscular malfunctioning, and psychological stress, which altogether facilitate the creation and perpetuation of myofascial trigger points. Contributing

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factors include chronic overload, constant static posture, injury, overtraining, postural deviation, biochemical abnormalities, and psychological distress [1].

Myofascial trigger points are common in the muscles of the cervical spine and shoulder region, including the upper trapezius, levator scapulae, scalene, and sternocleidomastoid muscles, which are especially vulnerable to mechanical stress and posture-related strain [2]. Patients with CMPS typically suffer from chronic neck pain, muscle stiffness, limited movement of the neck, and difficulties in performing daily activities, especially those involving long periods of neck movement [1]. These symptoms tend to worsen when exposed to prolonged postures, occupational pressure, cold weather, and mental stress [1].

The recent literature indicates an impressive correlation between the function and location of myofascial trigger points and classical acupuncture points (including Ashi points). It has been suggested that the majority of clinically applied trigger points coincide with Ashi points in Traditional Chinese Medicine, which can be considered to be tender palpation points with localized areas of heightened nociceptive sensitivity rather than fixed anatomical sites. Thus, this idea is based on the fact that both theories may be considered as alternative names for one and the same neurological-musculoskeletal phenomenon rather than independent concepts [3]. Moreover, several comparative studies in the field of anatomy and acupuncture suggest a significant similarity between the areas of pain distribution associated with trigger points and acupuncture meridians; it has been estimated that the level of correspondence can reach over 90% in certain anatomical structures [3, 4].

Cervical myofascial pain syndrome (CMPS) is a significant clinical problem and public health burden globally, given the high incidence of the disorder among patients with chronic neck pain. According to epidemiological research, an estimated 30% to 85% of all individuals globally suffering from chronic neck pain have myofascial trigger points in their neck muscles [4]. The global prevalence of cervical myofascial pain syndrome is estimated at around 10% to 20% in adults. Population-based epidemiological data for CMPS in Pakistan remain scarce. The syndrome is highly common amongst those individuals who have professions that involve repetitive use of their necks, for instance, office workers, dentists, and manual workers [5, 6]. Females have been reported to be more prone to develop the disorder than men, with a prevalence rate of 2:1 in females to males. CMPS predominantly occurs in those patients who belong to an age group between 30 and 60 years old [2, 7].

CMPS is mainly diagnosed by means of clinical approaches. In this context, identification of trigger points located in the muscles involved is considered the first step [8]. The standard procedure during the clinical assessment includes palpation for taut bands, local tenderness, and reproducing the patient's characteristic pain [1]. Other diagnostic techniques include pressure algometry for quantifying pressure pain thresholds, ultrasound and elastography for assessing muscle stiffness, and electromyography for detecting abnormal motor behavior [1].

Some of the methods that can be investigated in treating CMPS include medications, trigger point injection therapy, electrotherapy, laser therapy, stretching exercises, muscle strengthening exercises, posture correction techniques, manual therapy, i.e., myofascial release therapy and soft tissue manipulation, and psychological interventions dealing with psychosocial factors related to chronic pain [9–11]. While conservative multimodal therapy has long been considered the best modality for treating CMPS, there is yet no gold standard method of treatment for CMPS, indicating that further studies are required to investigate other methods of treatment.

Among the non-pharmacological methods of therapy, Dry Needling (DN) and Low-Level Laser Acupuncture (LLLA) are two popular techniques used in the treatment of CMPS [2, 3, 7, 12]. Dry Needling is considered to produce a good analgesic effect in cases of cervical myofascial pain syndrome [6]. On the other hand, Low-Level Laser Acupuncture (LLLA) is unique in its form as it represents an innovative approach that integrates the use of both acupuncture and Low-Level Laser therapy techniques. LLLA utilizes the peripheral and central pain mechanisms, thereby resulting in pain reduction and functional restoration in cervical myofascial pain disorders [3, 10].

Even though Dry Needling and Low-Level Laser Acupuncture are widely used clinically for their benefits, there is a huge gap in literature for the comparative effectiveness of both techniques in managing chronic CMPS. Additionally, there is insufficient literature about whether either treatment is better than the other, relating to the assessment of pain, function, physiological, and psychosocial parameters simultaneously, as most of the available literature is focused only on the assessment of pain outcomes, thus leaving insufficient information. Thus, this randomized controlled trial was formulated to explore and compare the clinical effects of Low-Level Laser Acupuncture and Dry Needling along with Routine Physical Therapy in patients with chronic cervical myofascial pain syndrome. The purpose of this study was to evaluate the changes in clinical outcomes (pain intensity, functional disability, cervical range of motion) and functional outcomes (health-related quality of life). It was hypothesized that both techniques would produce significant improvements clinically, but comparative variations in therapeutic outcomes will be observed because of the different physiological processes underlying each intervention.

By comparing the effectiveness of these two approaches objectively, this study intended to contribute to existing evidence for clinical practice.

Methodology

Study An Assessor-Blind, Randomized Controlled Trial ***(a parallel-group comparative superiority trial)*** was conducted in the Department of Physical Therapy at Rawal General & Dental Hospital, Pakistan. The study duration was 18 months, from June 2023 to November 2024. The study followed CONSORT guidelines for randomized trials to ensure methodological transparency and reproducibility.

Pilot study A pilot study was performed with 12 patients having Chronic Cervical Myofascial Pain Syndrome before the conduct of the main trial, in order to obtain preliminary outcome estimates and to assess some feasibility factors such as patient enrollment, patient selection, implementation of intervention, and outcome measurement variability. Besides, the pilot study was crucial for refining the research procedure and obtaining a reliable effect size estimate (Cohen's d≈0.83) used for sample size calculation.

Sample Size Calculation of sample size for the main RCT study was made with OpenEpi software using the formula: $n = \frac{\{(\delta_1^2+\delta_2^2)\times(Z_{1-\alpha/2}+Z_{1-\beta})^2\}}{|\mu_2-\mu_1|^2}$, where *n* represents the required sample size per group, δ_1^2 and δ_2^2 represent the variances of the two groups, $Z_{1-\alpha/2}$ corresponds to the selected significance level i.e. 5% margin of error with a 95% confidence interval, $Z_{1-\beta}$ corresponds to the desired statistical power i.e. 80% power, and $|\mu_2 - \mu_1|$ represents the expected difference between the group means. Post-intervention pain intensity (NPRS) means and variances were used as primary outcome values gained from the pilot study (Group A NPRS = 1.5 ± 1.049 and Group B NPRS = 2.33 ± 0.836), resulting in a minimum sample size of 42 participants. To strengthen the statistical power and reduce the risk of type II error, the sample size was doubled to 84, further inflated by 20% for anticipated attrition, thus resulting in a total sample size requirement of 100 subjects (50 per group). The same sample size was also deemed adequate to determine clinically significant differences in secondary variables such as NDI score and SF-36 quality of life score, based on previous literature in the same CMPS population [1, 2].

Sample A non-probability convenience sampling technique was employed to produce a study sample of 100 individuals with Chronic Cervical Myofascial Pain Syndrome due to the availability of eligible patients within the clinical setting and the time-bound nature of the study. The eligible participants presented with persistent symptoms for a minimum duration of three months. Participants exhibited unilateral or bilateral cervical pain and active myofascial trigger points in the upper trapezius, levator scapulae, and scalene muscles, identified through palpation of a taut band and a positive jump sign. Other requirements for admission were: moderate neck disability [6] (NDI score between 21% and 40%), NPRS score greater than 3 [2], and a cervical range of motion that was limited by up to 60% [6]. People who had bleeding disorders, were on anticoagulants, had cervical spine surgery in the past year, had other treatments in the past three months, or people with spinal deformities or serious medical problems were excluded. All participants were asked to sign the informed consent form after being informed of the nature of the methods and procedures to be followed.

Although convenience sampling was used for recruitment, the allocation of participants into groups followed a rigorous random allocation approach to minimize selection bias. This distinction between recruitment method and allocation procedure has been maintained throughout the study design.

Randomization procedure

Randomization was performed using a *Random Allocation Software (version 2)* through which a computer-generated random sequence *(Block Randomization Method)* was used to ensure balanced group allocation. The allocation sequence was developed by an independent researcher who had no involvement in either participant recruitment, outcome measurement, or intervention administration, thereby maintaining allocation concealment.

Group assignments were placed in sequentially numbered, opaque, sealed envelopes (SNOSE technique), which were opened only after completion of baseline assessments to ensure allocation concealment and prevention of selection bias.

Group A 50 patients, 29 women and 21 men, with an average age of 34.24±7.68 years, had Low-Level Laser Acupuncture along with Routine Physical Therapy.

Group B 50 patients, 27 women and 23 men, with an average age of 33.10±6.74 years, had Dry Needling along with Routine Physical Therapy.

Blinding scheme

Considering the characteristics of the interventions used in the study (LLLA and Dry Needling), it is not possible

to blind participants as well as treating physiotherapists because the nature of such treatments makes them distinctive. Thus, to minimize any detection or assessment bias associated with the research design and process, the following blinding strategy was adopted: (i) a single-assessor blind trial design was employed; (ii) the outcome assessor was blinded to group allocation; (iii) participants were asked not to reveal treatment allocation when being assessed after the treatment; and (iv) data were analyzed by coding groups as A and B. The method of assessor blinding was adopted in order to increase the internal validity of the study, despite the constraints that come with a manual therapy approach-based intervention.

Ethical Approval and Trial Registration This randomized controlled trial was conducted in accordance with the ethical principles of the *Declaration of Helsinki for research involving human participants*. Ethical approval for the study was obtained from the Institutional Review Board/Ethics Committee of the University of Lahore, ensuring compliance with established ethical principles and participant protection. ***(REC No. 443, approved on 1ST December 2022)*** All participants provided written informed consent before enrollment in the study. The clinical trial was prospectively registered in the ***Iranian Registry of Clinical Trials (IRCT)*** under the *Registration No.* ***(IRCT20190618043930N2)***. The registry record is publicly accessible through the IRCT database.

Intervention procedure

Both groups had received ***Routine Physical Therapy (RPT)*** before applying comparable protocols as a part of standard treatment. The patient was educated about the correct neck posture through practical demonstration.

The C1-C7 area of the cervical spine was treated with hot fermentation for about 10 min. The Hold- relax technique with sustained stretch was performed on the Scalene, Levator scapulae, and Upper Trapezius muscles on the affected side in 03 sets of 05 repetitions each with a 15-second hold. Low to moderate isometric strengthening exercises of the upper cervical muscles in 2 sets of 10 repetitions in each direction (flexion, extension, side bending, and rotation).

Group A Intervention Participants assigned to the ***Low-Level Laser Acupuncture (LLLA)*** group were provided with protective goggles to ensure ocular safety during each treatment session. The intervention utilized an infrared Class IIIb Gallium-Aluminum-Arsenide (Ga-Al-As) diode laser device, a modality well-documented for its biostimulatory and analgesic properties in musculoskeletal pain management [11]. The laser device (830 nm, near-infrared) was applied in pulsed mode at a frequency of 9.12 Hz, with a spot diameter of 1 cm (spot area: 0.785 cm^2). The absolute power output was 78.5 mW, corresponding to a power density (fluence) of 100 mW/cm^2 [12]. An energy density (irradiance) of 5 J/cm^2 was delivered to each point over an irradiation time of approximately 50 s, yielding an energy of 3.93 J per point. 10–11 acupuncture and myofascial trigger points were irradiated per session, resulting in a total session energy of 39.3 J and a total irradiation time of approximately 500 s (~ 8.3 min). The laser probe was applied in direct contact with and perpendicular to the skin surface overlying the ipsilateral site of maximal pain [3, 12, 13]. This precise protocol ensured standardized energy delivery to the target tissues, promoting local photobiomodulation effects such as enhanced microcirculation, reduction of inflammatory mediators, and modulation of nociceptive signaling pathways.

The ***acupuncture points*** used were Shousanli (Large Intestine Meridians, LI 10): Hegu (Large Intestine Meridians, LI 4), Waiguan (Sanjiao Meridians, TE 5), and Houxi (Small Intestine Meridians, SI 3). They were anatomically fixed and applied uniformly across participants [3].

The ***trigger points*** on the Scalene, Levator scapulae, and upper Trapezius muscles were identified using standard clinical criteria, including palpable taut bands, local tenderness, and reproduction of referred pain, and they were patient-specific [9].

All these points were marked with an 'X' mark. A detailed description of these points with their anatomical location, depth, and identification method is shown in Table 1.

Group B Intervention In ***Dry needling (DN)*** group, while the patient is in prone position, the therapist wore the surgical gloves, sterilized the area with alcohol pad and used standard clinical criteria to identify the ***trigger points*** (Scalene, Levator scapulae and upper Trapezius muscles) and ***acupuncture points*** (Shousanli (Large Intestine Meridians, LI 10): Hegu (Large Intestine Meridians, LI 4), Waiguan (Sanjiao Meridians, TE 5), and Houxi (Small Intestine Meridians, SI 3). All these points were marked as 'X'. A sterile single-use disposable filiform needle (25G acupuncture needle 0.3* 30 mm) was placed on the trigger points as well as on the acupoints to a depth of 10–20 mm [14]. The needles were manipulated to elicit local twitch responses/De Qi sensations and retained for approximately 30–60 s per point, depending upon the patient's tolerance [2, 4]. The needles were removed, and any bleeding that occurred was controlled. The procedure was performed under strict aseptic precautions to ensure

Table 1 Shows the anatomical location & identification methods of Acupuncture points and Trigger points, along with LLLA & DN application techniques used in the trial

Structure/Point	Anatomical location & identification	Dry needling application	LLLA application
LI 10 — Shousanli (Large Intestine Meridian)	Located 2 inches distal to LI 11 on the dorsal-radial forearm over extensor carpi radialis brevis. Identified with elbow flexed to 90° and measured two finger-breadths distal to the lateral epicondyle crease.	Sterile filiform needle 25G (0.3 mm × 30 mm) inserted perpendicular to 10–20 mm depth for 30–60 s. Needle manipulated until De Qi sensation is elicited.	Infrared diode Laser probe 1 cm in diameter applied perpendicular to the skin surface with light contact. *Parameters*: 830 nm wavelength, power output of 100 mW/cm², pulsed frequency of 9.12 Hz, 5 J energy density, 50 s irradiation on each point.
LI 4 — Hegu (Large Intestine Meridian)	Located on the dorsum of the hand in the first dorsal interosseous muscle at the midpoint of the 2nd metacarpal radial border. Identified by approximating thumb crease to the web space between thumb and index finger.	Needle inserted obliquely toward the 2nd metacarpal to a depth of 10–15 mm with De Qi stimulation for 30–60 s.	
TE 5 — Waiguan (Sanjiao Meridian)	Located on the dorsal forearm, 2 inches proximal to the dorsal wrist crease between the radius and ulna over extensor digitorum.	Perpendicular dry needling performed to 10–20 mm depth with controlled manipulation for 30–60 s.	
SI 3 — Houxi (Small Intestine Meridian)	Located on the ulnar border of the hand proximal to the 5th metacarpophalangeal joint at the junction of red and white skin when the hand is lightly clenched.	Perpendicular insertion into interosseous tissue to 5–10 mm depth with elicitation of De Qi sensation for 30–60 s.	
Upper Trapezius MTrP1 & MTrP2	*MTrP1* located midway between acromion and C7 spinous process in the upper trapezius belly. *MTrP2* positioned more laterally near the acromioclavicular region. Identified via pincer palpation. *Depth*: Superficial to moderate, around 10–15 mm.	MTrP1: Pincer grip technique with perpendicular/oblique insertion, 5–10 mm depth for 30–60 s. MTrP2: Flat palpation with 3–7 mm insertion depth for 30–60 s. Local twitch response used for confirmation.	Probe positioned perpendicular over each trigger point with no compression of taut bands. *Parameters*: 830 nm wavelength, power output of 100 mW/cm², frequency of 9.12 Hz, 5 J energy density, 50 s irradiation.
Levator Scapulae Primary & Secondary MTrPs	*Primary MTrP* Located at the angle of the neck near C3–C4, approximately 2–3 cm lateral to the C4 spinous process beneath trapezius. *Secondary MTrP* near superior medial scapular border. *Depth*: Deep; requires firm, direct, or "hooking" pressure.	Flat palpation technique used. Needle directed superiorly for 30–60 s along muscle fibers to 15–25 mm depth through trapezius while patient rotates head away from treated side.	Laser probe applied over the C3–C4 angle-of-neck site with transcutaneous photon penetration reaching deeper tissues safely.
Scalene Group Anterior, Middle & Posterior Scalene MTrPs	*Anterior scalene MTrP* located lateral to SCM at C4–C5 between SCM clavicular head and external jugular vein. Identified in patient positioned supine or semi-reclined with the head rotated approximately 30° away from the treated side. *Middle scalene MTrP* positioned posteriorly between C4–C6. Identified in patient supine with slight contralateral cervical rotation. *Posterior Scalene MTrP* located in the posterior scalene muscle belly, posterior to the middle scalene and anterior to the upper trapezius, usually at the C5–C7 cervical level near the lateral neck. Identified in patient seated or supine with mild contralateral side flexion. *Depth*: Deep in the neck, behind SCM.	Needle inserted 5–12 mm maximum depth for 30–60 s with strict superior – medial angulation away from lung apex. Extreme caution required.	Probe applied lateral to SCM. Considered the safer modality due to proximity of brachial plexus, vessels, and lung apex.

participant safety. All used needles were disposed of in accordance with institutional biomedical waste disposal protocols [2, 4, 14].

Low-Level Laser Acupuncture and Dry Needling were applied to both clinically identified myofascial trigger points and selected acupuncture points shown in Table 1 due to the documented overlap between trigger point locations and Ashi/acupuncture points, as well as their shared role in neuromodulation of pain pathways i.e. MTrP needling or laser irradiation addresses the peripheral myofascial pathology (dysfunctional motor endplate, local nociceptive sensitisation) while acupuncture point stimulation provides segmental and supraspinal neuromodulation [3, 4].

All interventions were delivered by a single registered Physical Therapist trained in musculoskeletal and trigger/acupuncture point–based therapies, following standardized protocols to ensure consistency. *Treatment fidelity* was

monitored through session checklists completed for every treatment encounter, documenting point selection, needle depth or laser parameters, treatment duration, and patient response. Fidelity checklists were reviewed monthly by the principal investigator.

Measurable outcome variables

Primary outcome measure Pain intensity.

Secondary outcomes Neck Functional disability, Cervical range of motion, and health-related quality of life.

The Numeric Pain Rating Scale (NPRS) was used to rate patients' pain intensity on a scale from 0 (no pain) to 10 (the worst imaginable pain [2]. The Neck Disability Index (NDI) is a 10-item questionnaire that assesses the impact of neck pain on daily activities, including personal care, lifting, reading, work, driving, sleeping, and recreation. Each item was scored from 0 to 5, with higher scores indicating greater disability [6]. A goniometer was used to assess cervical range of motion (ROM) in flexion, extension, lateral flexion, and rotation. In CMPS, it helps evaluate mobility limitations due to myofascial trigger points [1]. The SF-36 item survey with a 0–100 score range was used, with higher scores indicating better health status. It is a RAND Short Form 36-item Health Survey Questionnaire measuring health-related quality of life across eight domains: physical functioning, role limitations due to physical health, bodily pain, general health perceptions, vitality, social functioning, role limitations due to emotional problems, and mental health [5]. The validity, reliability, specificity, and sensitivity of all these outcome measures were described in Table 2 [1, 2, 5, 6].

Outcome assessment timeline

The treatment procedures were repeated for 3 sessions per week for 6 weeks (18 sessions). The Pre-test, Mid-test & Post-test scores of NPRS, NDI, Goniometer and SF-36 Health Questionnaire were collected on the first, ninth, and eighteenth treatment sessions, respectively, and were analyzed by an outcome Assessor, who was kept blind throughout the sessions to reduce any bias in the study.

Follow-up assessment period

At the end of the intervention phase, participants had to undergo a four-week follow-up phase during which no further interventions were made. The objective of the follow-up phase was the determination of the short-term sustainability of the treatment effect after the discontinuation of therapy.

In the course of the follow-up phase, participants were asked to continue their normal routine activity; however, they were not supposed to use any further physiotherapy or any other invasive procedure related to the neck pain. The implementation of the guidelines was ensured based on the participants' self-reports.

Assessment of the outcomes was carried out at the end of the four weeks of follow-up to find out how the treatment effect persisted. The variables that were to be measured were the following: pain intensity (Numeric Pain Rating Scale), functional disability (Neck Disability Index), cervical ROM (goniometer), and health-related quality of life (Short Form-36 Health Survey). The participants were contacted and invited to the follow-up visit at which all the outcomes mentioned above were to be recorded.

Data analysis

All data analyses were conducted according to the *intention-to-treat* principle using IBM SPSS Statistics version 26.0. Before applying inferential statistics, data normality was examined through the Shapiro–Wilk test, which indicated a non-normal distribution ($p < 0.05$, 95% CI). Consequently, non-parametric statistical procedures were employed. Descriptive statistics were presented as median values with corresponding interquartile ranges (25th and 75th percentiles) to represent data dispersion (Table 3). For *intergroup comparisons*, the Mann–Whitney U test was utilized to determine statistically significant differences in clinical and functional outcomes. In addition to statistical significance, effect size analysis (r) was conducted to determine the magnitude and clinical relevance of the observed differences between groups. *Intragroup analyses* across multiple time intervals were performed using the Friedman test (the non-parametric equivalent of repeated-measures ANOVA) to identify statistically significant temporal changes (Pre-test, Mid-test, and Post-test) within each group. A two-tailed significance threshold of $p < 0.05$ with a 95% confidence

Table 2 Reliability, validity, sensitivity, and specificity of the study measures

Instrument	Reliability (ICC)	Validity	MCID	Specificity	Application in CMPS
NPRS	0.95–0.96	High	~2 points	Moderate	Pain intensity assessment
NDI	0.91	High	~7.5 points	Moderate	Functional disability evaluation
Goniometer	≥0.986	High	≤5.23%	Low–Moderate	Cervical ROM measurement
SF-36	0.85–0.87	High	Varies by domain	Low	Quality of life assessment

Table 3 Descriptive statistics of study participants

Variables	Time Point	Group A Median (IQR)	Group B Median (IQR)
Age	Baseline	33.5 (28.2, 38.7)	30 (26, 34)
NPRS	Pre-Test	7 (6, 8)	7 (6, 8)
	Mid-Test	5 (4.5, 5.5)	5 (4.5, 5.5)
	Post- Test	1.78 (1.2, 2.2)	3 (2.5, 3.5)
NDI	Pre-Test	35 (31, 39)	34 (30.5, 37.5)
	Mid-Test	23 (19, 27)	22 (19.3, 24.6)
	Post- Test	10 (8.5, 11.5)	16 (13, 19)
SF-36 Health Domains			
Role Limitations (Physical)	Pre-Test	25 (20.5, 29.5)	26 (20.5, 31.5)
	Mid-Test	34 (29.5, 38.5)	34 (28.3, 39.6)
	Post- Test	50 (45, 55)	42 (35.3, 48.6)
Physical Functioning	Pre-Test	18 (15.5, 20.5)	20 (16.2, 23.7)
	Mid-Test	32.5 (29.5, 35.5)	28.5 (24.5, 32.5)
	Post- Test	50 (47.5, 52.5)	43 (38.3, 47.6)
Bodily Pain	Pre-Test	20.5 (16, 25)	24.5 (19.8, 29.1)
	Mid-Test	31 (26.5, 35.5)	35 (30.2, 39.7)
	Post- Test	48 (43.5, 52.5)	50 (44.8, 55.1)
General Health	Pre-Test	55 (47.5, 62.5)	53 (48.5, 57.5)
	Mid-Test	61 (54, 68)	60 (55.3, 64.6)
	Post- Test	68 (61, 75)	67 (62, 72)
Vitality	Pre-Test	42 (36.5, 47.5)	43 (39, 47)
	Mid-Test	48 (42.5, 53.5)	49 (44.8, 53.1)
	Post- Test	57 (52.5, 61.5)	61 (57.1, 64.8)
Social Functioning	Pre-Test	33.5 (28.5, 38.5)	51 (47.3, 54.6)
	Mid-Test	51.5 (46.5, 56.5)	60 (56, 64)
	Post- Test	73 (68.5, 77.5)	73.5 (69.7, 77.2)
Role Limitations (Emotional)	Pre-Test	49 (44, 54)	62.5 (55, 70)
	Mid-Test	59.5 (54.5, 64.5)	70 (62.8, 77.1)
	Post- Test	68 (63, 73)	74 (66.8, 81.1)
Mental Health	Pre-Test	43.5 (38, 49)	48 (43, 53)
	Mid-Test	51 (45.5, 56.5)	53 (48, 58)
	Post- Test	59 (54, 64)	60 (55.7, 64.2)
Cervical ROM			
Flexion	Pre-Test	44.4 (39.9, 48.9)	47 (42.3, 51.6)
	Mid-Test	46.5 (42.2, 50.7)	48.5 (43.8, 53.1)
	Post- Test	47.5 (43.2, 51.7)	49.5 (44.8, 54.1)
Extension	Pre-Test	51 (46, 56)	52 (49.8, 54.1)
	Mid-Test	53 (48, 58)	53 (50.8, 55.1)
	Post- Test	55 (49.5, 60.5)	55 (52.3, 57.6)
Ipsilateral Lateral Flexion	Pre-Test	43 (40.5, 45.5)	43 (40.8, 45.1)
	Mid-Test	46 (43.3, 48.6)	45 (42.5, 47.5)
	Post- Test	51 (48.5, 53.5)	47 (44.3, 49.6)
Contralateral Lateral Flexion	Pre-Test	42 (39, 45)	38 (34, 42)
	Mid-Test	45 (42.3, 47.6)	41 (37, 45)
	Post- Test	49 (46.5, 51.5)	43 (38.8, 47.1)
Ipsilateral Rotation	Pre-Test	75.5 (72, 79)	74 (70.3, 77.6)
	Mid-Test	77 (73.5, 80.5)	76 (72.3, 79.6)
	Post- Test	80 (76.7, 83.2)	78.5 (74.8, 82.1)
Contralateral Rotation	Pre-Test	76 (70.8, 81.1)	75.5 (72, 79)
	Mid-Test	78 (72.8, 83.1)	77 (73.3, 80.6)
	Post- Test	80 (74.5, 85.5)	78 (74.2, 81.7)

interval was established to accept or reject the null hypothesis. When significant differences were detected, Wilcoxon signed-rank tests with Bonferroni adjustment ($\alpha = 0.017$) were performed for pairwise post-hoc comparisons and to further evaluate the magnitude and direction of change in pain intensity, neck disability, cervical range of motion (C-ROM), and health-related quality of life as measured by the SF-36 questionnaire.

Results

The screening of 115 patients with clinically diagnosed chronic CMPS was done for possible participation in the study. After exclusion of 15 participants, 100 fulfilled the inclusion criteria of the study. The 50 participants were allocated to Group A and the 50 participants to Group B. During the study, 09 participants dropped out, i.e., Group A: 2 after pretest and 2 after midtest, and Group B: 3 after pretest and 2 after midtest. A total of 91 participants successfully completed the 6-week follow-up period.

Statistical handling of dropouts

To address the missing data, the Last Observation Carried Forward (LOCF) method was employed as the primary imputation approach under the assumption that data were Missing Completely at Random (MCAR), in order to preserve the integrity of the randomized design and intention-to-treat principle, whereby the most recent available measurement from the previous assessment was used as the final value, maintaining the group comparability after randomization. The use of LOCF was considered appropriate because the attrition rate remained relatively low (9%, 9/100), thereby minimizing the potential influence of missing observations on overall treatment results. This approach also reduced the risk of bias associated with complete-case analysis and helped preserve statistical power for intergroup and intra-group comparisons. The overall flow of participant recruitment, allocation, follow-up, and analysis is presented in accordance with the CONSORT (Consolidated Standards of Reporting Trials) guidelines (Fig. 1).

Intragroup analyses

The ***Friedman test*** results for Groups A and B, shown in Tables 4 and 5, respectively, indicated a statistically significant effect of time on all clinical and functional variables evaluated *(χ^2 values ranging from 67.8 to 94.1, $p < 0.001$)* and (χ^2 values ranging from 72.8 to 97.3, $p < 0.001$),

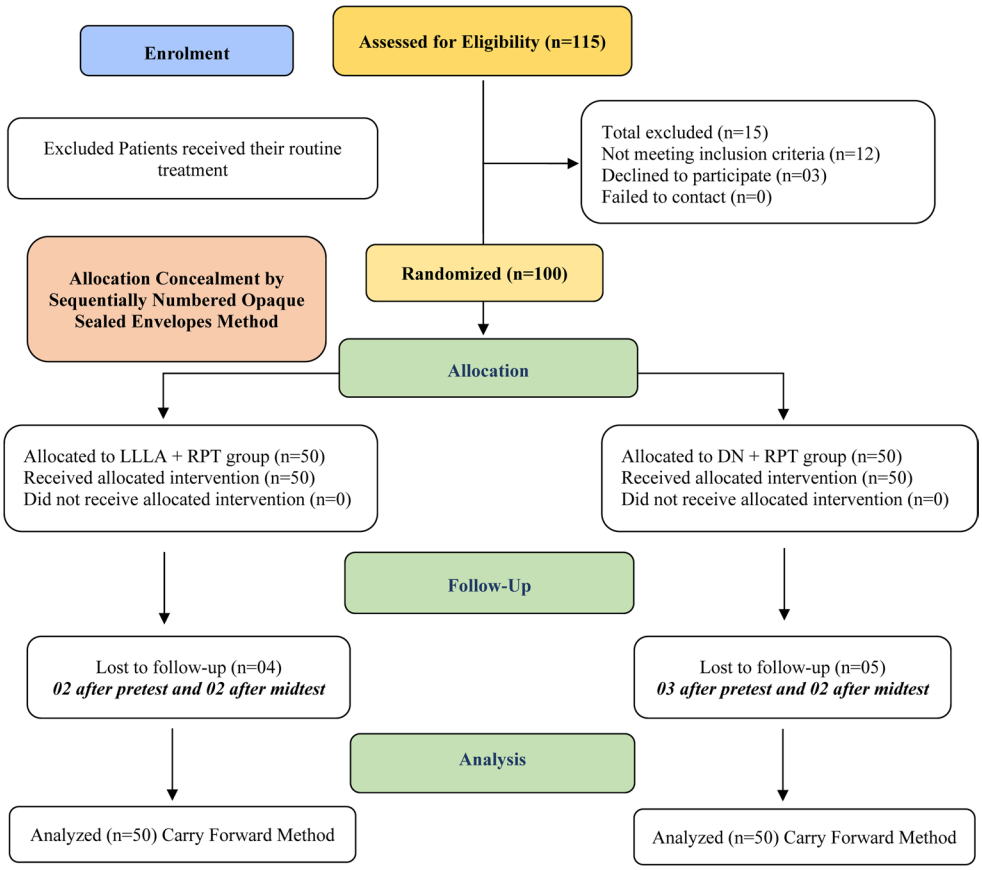


Fig. 1 CONSORT diagram and study flow chart

Table 4 Group A intragroup analyses: Friedman test (χ^2), Post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni adjustment ($\alpha = 0.017$)

Group A outcome variable	Friedman χ^2	*p*-value (95% CI)	Pre – Mid (Z)	*p*-value (95% CI)	Pre – Post (Z)	*p*-value (95% CI)	Mid – Post (Z)	*p*-value (95% CI)	Bonferroni corrected significance
NPRS	92.4	<0.001	−6.84	<0.001	−8.52	<0.001	−5.71	<0.001	Significant ($p<0.017$)
NDI	94.1	<0.001	−6.76	<0.001	−8.63	<0.001	−5.89	<0.001	Significant
SF-36 Role Limitations (Physical)	88.3	<0.001	−6.21	<0.001	−8.11	<0.001	−5.02	<0.001	Significant
Physical Functioning	86.5	<0.001	−6.14	<0.001	−8.02	<0.001	−4.97	<0.001	Significant
Bodily Pain	90.6	<0.001	−6.63	<0.001	−8.44	<0.001	−5.40	<0.001	Significant
General Health	84.9	<0.001	−5.94	<0.001	−7.83	<0.001	−4.75	<0.001	Significant
Vitality	82.7	<0.001	−5.82	<0.001	−7.71	<0.001	−4.62	<0.001	Significant
Social Functioning	80.4	<0.001	−5.65	<0.001	−7.56	<0.001	−4.51	<0.001	Significant
Role Limitations (Emotional)	79.8	<0.001	−5.59	<0.001	−7.44	<0.001	−4.43	<0.001	Significant
Mental Health	78.1	<0.001	−5.47	<0.001	−7.33	<0.001	−4.36	<0.001	Significant
Cervical Flexion	75.9	<0.001	−5.22	<0.001	−7.01	<0.001	−4.05	<0.001	Significant
Cervical Extension	73.6	<0.001	−5.09	<0.001	−6.87	<0.001	−3.96	<0.001	Significant
Ipsilateral Lateral Flexion	71.8	<0.001	−4.96	<0.001	−6.74	<0.001	−3.88	<0.001	Significant
Contralateral Lateral Flexion	70.4	<0.001	−4.88	<0.001	−6.66	<0.001	−3.82	<0.001	Significant
Ipsilateral Rotation	69.1	<0.001	−4.74	<0.001	−6.51	<0.001	−3.70	<0.001	Significant
Contralateral Rotation	67.8	<0.001	−4.65	<0.001	−6.43	<0.001	−3.63	<0.001	Significant

Values are reported as Z statistics with corresponding *p*-values

Table 5 Group B intragroup analyses: Friedman test (χ^2), Post-hoc pairwise comparisons using the Wilcoxon signed-rank test with Bonferroni adjustment ($\alpha = 0.017$)

Group B Outcome Variable	Friedman χ^2	*p*-value (95% CI)	Pre – Mid (Z)	*p*-value (95% CI)	Pre – Post (Z)	*p*-value (95% CI)	Mid – Post (Z)	*p*-value (95% CI)	Bonferroni Corrected Significance
NPRS	97.3	< 0.001	−7.02	< 0.001	−8.91	< 0.001	−6.11	< 0.001	Significant ($p < 0.017$)
NDI	95.8	< 0.001	−6.94	< 0.001	−8.77	< 0.001	−5.94	< 0.001	Significant
SF-36 Role Limitations (Physical)	90.7	< 0.001	−6.52	< 0.001	−8.41	< 0.001	−5.44	< 0.001	Significant
Physical Functioning	88.6	< 0.001	−6.34	< 0.001	−8.24	< 0.001	−5.21	< 0.001	Significant
Bodily Pain	92.1	< 0.001	−6.73	< 0.001	−8.65	< 0.001	−5.67	< 0.001	Significant
General Health	87.5	< 0.001	−6.18	< 0.001	−8.02	< 0.001	−5.09	< 0.001	Significant
Vitality	85.3	< 0.001	−6.04	< 0.001	−7.86	< 0.001	−4.94	< 0.001	Significant
Social Functioning	83.9	< 0.001	−5.93	< 0.001	−7.72	< 0.001	−4.82	< 0.001	Significant
Role Limitations (Emotional)	82.1	< 0.001	−5.81	< 0.001	−7.61	< 0.001	−4.69	< 0.001	Significant
Mental Health	80.4	< 0.001	−5.70	< 0.001	−7.49	< 0.001	−4.56	< 0.001	Significant
Cervical Flexion	78.2	< 0.001	−5.52	< 0.001	−7.28	< 0.001	−4.31	< 0.001	Significant
Cervical Extension	76.7	< 0.001	−5.39	< 0.001	−7.12	< 0.001	−4.18	< 0.001	Significant
Ipsilateral Lateral Flexion	75.3	< 0.001	−5.25	< 0.001	−6.98	< 0.001	−4.07	< 0.001	Significant
Contralateral Lateral Flexion	74.6	< 0.001	−5.19	< 0.001	−6.92	< 0.001	−3.99	< 0.001	Significant
Ipsilateral Rotation	73.4	< 0.001	−5.06	< 0.001	−6.78	< 0.001	−3.90	< 0.001	Significant
Contralateral Rotation	72.8	< 0.001	−4.98	< 0.001	−6.70	< 0.001	−3.82	< 0.001	Significant

Values are reported as Z statistics with corresponding *p*-values. All post-hoc Z values negative, reflecting improvement (score reduction for NPRS/NDI; score increase for SF-36/ROM)

respectively. All three pairwise post-hoc comparisons by the Wilcoxon Signed rank test (Pre→Mid, Pre→Post, and Mid→Post) survived Bonferroni correction *(corrected $\alpha = 0.017$)* for both groups.

Effect size (r) interpretation

The *intra-group effect size analysis*, as shown in Fig. 2, indicates that both Group A and Group B showed statistically significant and clinically meaningful improvements across all outcome variables over time, with all pairwise comparisons remaining significant at $p < 0.001$ (95% CI). However, Group B exhibited consistently larger early effect sizes than Group A across pain, disability, quality-of-life, and cervical mobility outcomes, suggesting a comparatively better early response to treatment.

Intergroup analyses

At *baseline*, the ***Mann–Whitney U statistic*** demonstrated a statistically insignificant difference ($p > 0.05$, 95% CI), with almost all the outcome variables, confirming that the groups were comparable. As the treatment progressed, however, variations became prominent across all the variables in both groups, as shown in Table 6. The inter-group effect sizes (r) for pain intensity, functional disability, health-related quality of life (SF-36 domains), and cervical range of motion variables over the study timeline are shown in Fig. 3.

Discussion

The present study explored and compared the effectiveness of Low-Level Laser Acupuncture (LLLA) and Dry Needling (DN) along with Routine Physical Therapy (RPT) in patients with Chronic Cervical Myofascial Pain Syndrome (chronic CMPS). Non-parametric statistical analyses were used to analyze the within-group and between-group changes across four clinical domains, i.e., pain intensity, functional disability, cervical range of motion, and health-related quality of life at three assessment timepoints: pretest, midtest, and posttest. *Intra-group comparisons* were conducted through Friedman tests, followed by post hoc pairwise analyses by Wilcoxon signed-rank tests and Bonferroni correction (corrected $\alpha = 0.017$). In contrast, *inter-group differences* were determined via the Mann–Whitney U test with effect sizes reported as the rank-biserial correlation coefficient r. Since both groups underwent the same RPT protocol concurrently, and there was no placebo or sham control included, the findings must be interpreted as a comparative assessment of LLLA and DN within multimodal rehabilitation rather than the exclusive effectiveness of each technique alone.

Both interventions produced statistically significant, clinically meaningful *within-group improvements* across all 16 outcome variables at every measurement transition (Pre→Mid, Mid→Post, and Pre→Post), with all Friedman tests reaching $p < 0.001$ (95% CI) and all 48 post-hoc

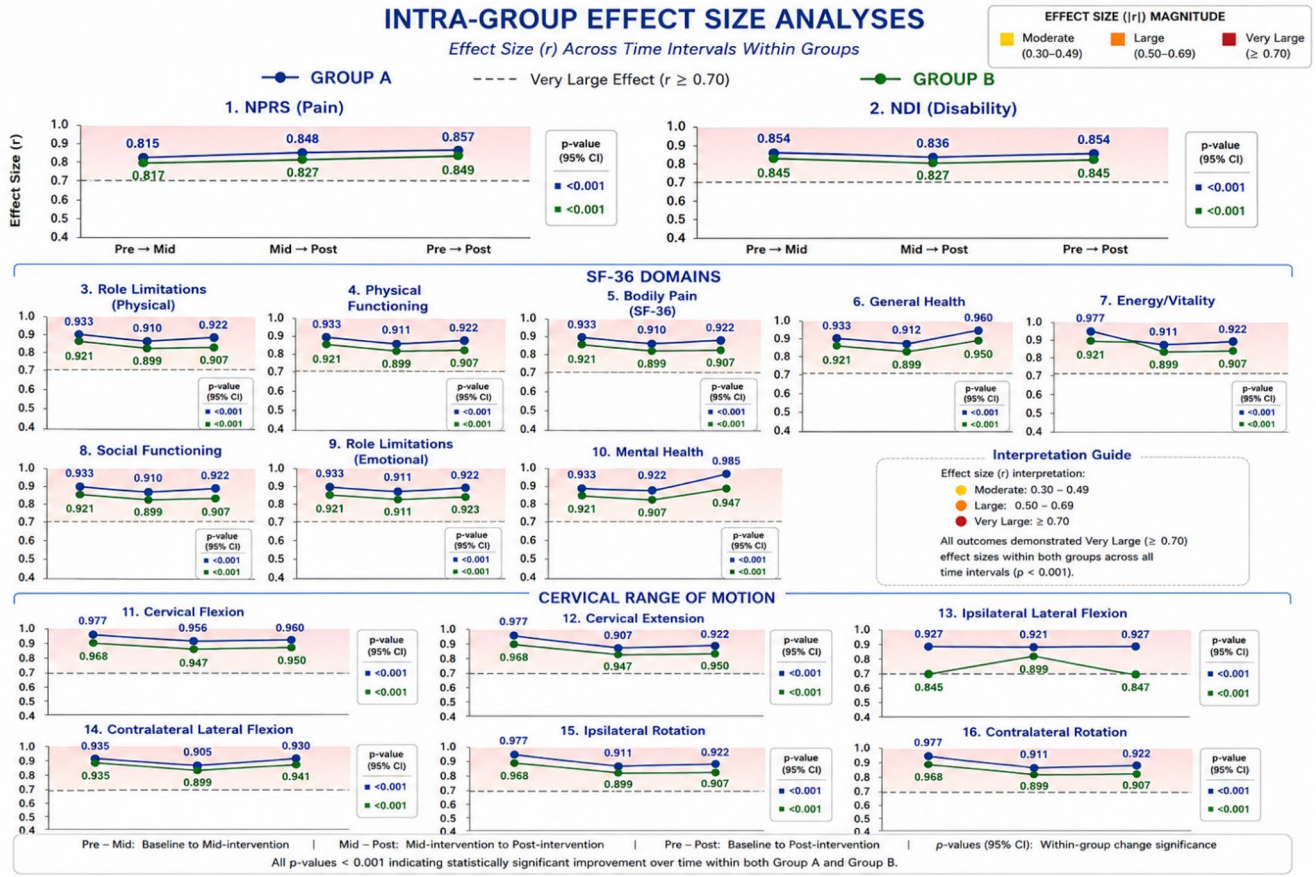


Fig. 2 Illustrates the temporal analyses of intragroup effect sizes with statistically significant differences ($p < 0.001$, 95% CI)

comparisons surviving Bonferroni correction in both groups. Within-group effect sizes derived from the Wilcoxon analysis ranged from $r = 0.815$ to $r = 0.985$ across all outcomes and transitions, values that substantially exceed Cohen's threshold for a large effect ($r = 0.50$) and sit within what may be characterised as the very large to exceptional range, confirming the clinical significance of both interventions across all measured outcomes. Although both interventions were effective, they differed notably in the timing and pattern of improvement over time. The *between-group* Mann-Whitney U analysis revealed that Group A (LLLA + RPT) produced progressively larger intergroup advantages as treatment advanced, with effect sizes growing monotonically from trivial or small at pretest to large at posttest across primary outcomes, a pattern consistent with the cumulative, dose-dependent mechanism of photobiomodulation. Group B (DN + RPT), by contrast, demonstrated higher Friedman χ^2 values and larger Pre→Mid (Z) scores across pain, disability, and ROM compared to Group A, reflecting the neurophysiologically rapid early analgesic response characteristic of trigger point deactivation. This temporal contrast, early neurophysiological gains for DN + RPT and progressive cumulative gains for LLLA + RPT, constitutes the central clinical finding of this trial and has direct implications for intervention selection, treatment sequencing, and patient-centred goal setting in the management of chronic CMPS.

In the present study, the Last Observation Carried Forward (LOCF) method was used to manage missing data arising from nine participant withdrawals (9%, 9/100), under the assumption that data were Missing Completely at Random (MCAR). All nine withdrawals were documented as occurring for reasons unrelated to treatment outcome or adverse events, supporting the MCAR assumption. At a dropout rate of 9%, the volume of missing data falls substantially below the threshold at which the imputation method becomes a material determinant of trial conclusions; with fewer than one in ten observations imputed, no standard alternative approach, including multiple imputation or mixed-model repeated-measures analysis, would be expected to produce meaningfully different results. Furthermore, in the context of a progressive rehabilitation intervention where both groups demonstrated continuous improvement across all timepoints, LOCF is inherently conservative: carrying the last observed value forward underestimates the true magnitude of improvement rather than inflating it, meaning any statistically significant findings are more likely to represent

Table 6 Inter-group comparisons (Mann–Whitney U test) with effect size (r): Trivial < 0.1, Small 0.1–0.3, Medium 0.3–0.5, Large 0.5–0.7, Very large > 0.7, p*= significant (< 0.05), A > B = Group A superior to Group B, ⚑ = Pretest imbalance — interpret posttest difference with caution. Convergence = groups not significantly different by posttest despite earlier divergence

OUTCOME	PRETEST				MIDTEST				POSTTEST			
	U	*r*	*p*-value (95% CI)	Group Effect	U	*r*	*p*-value (95% CI)	Group Effect	U	*r*	*p*-value (95% CI)	Group Effect
NPRS	1206	0.031	0.757	Equal	963.5	0.209	0.037*	A > B	457	0.561	<0.001*	A > B
NDI	1063.5	0.129	0.197	Equal	1061	0.131	0.191	Equal	471.5	0.539	<0.001*	A > B
SF-36 DOMAINS												
Role Limitations (Physical)	1050.5	0.138	0.168	Equal	1163	0.06	0.548	Equal	722	0.364	<0.001*	A > B
Physical Functioning	1027.5	0.154	0.123	Equal	826	0.293	0.003*	A > B	697.5	0.381	<0.001*	A > B
Bodily Pain	788	0.319	0.001*	A > B	785.5	0.321	0.001*	A > B	946	0.21	0.036*	A > B
General Health	1098	0.105	0.294	Equal	1151.5	0.068	0.496	Equal	1077	0.119	0.232	Equal
Vitality	1037.5	0.147	0.142	Equal	1072.5	0.123	0.22	Equal	921	0.227	0.023*	A > B
Social Functioning	218.5	0.712	<0.001*	A > B	421.5	0.572	<0.001*	A > B	1058.5	0.132	0.186	Converged
Role Limitations (Emotional)	431	0.565	<0.001*	A > B	521.5	0.503	<0.001*	A > B	712.5	0.371	<0.001*	A > B
Mental Health	802.5	0.309	0.002*	A > B	1057	0.133	0.182	Converged	1087.5	0.112	0.261	Converged
CERVICAL ROM												
Flexion	993.5	0.177	0.077	Equal	1002.5	0.171	0.087	Equal	1007.5	0.167	0.094	Equal
Extension	1161	0.062	0.539	Equal	1161	0.062	0.539	Equal	1139.5	0.076	0.445	Equal
Ipsi. Lateral Flexion	1120	0.09	0.367	Equal	848.5	0.278	0.005*	A > B	571	0.47	<0.001*	A > B
Contra. Lateral Flexion	853.5	0.274	0.006*	A > B ⚑	856	0.272	0.006*	A > B	666.5	0.403	<0.001*	A > B
Ipsi. Rotation	1180.5	0.048	0.631	Equal	1173.5	0.053	0.597	Equal	1059	0.132	0.187	Equal
Contra. Rotation	1120.5	0.09	0.371	Equal	1115.5	0.093	0.353	Equal	1027.5	0.154	0.124	Equal

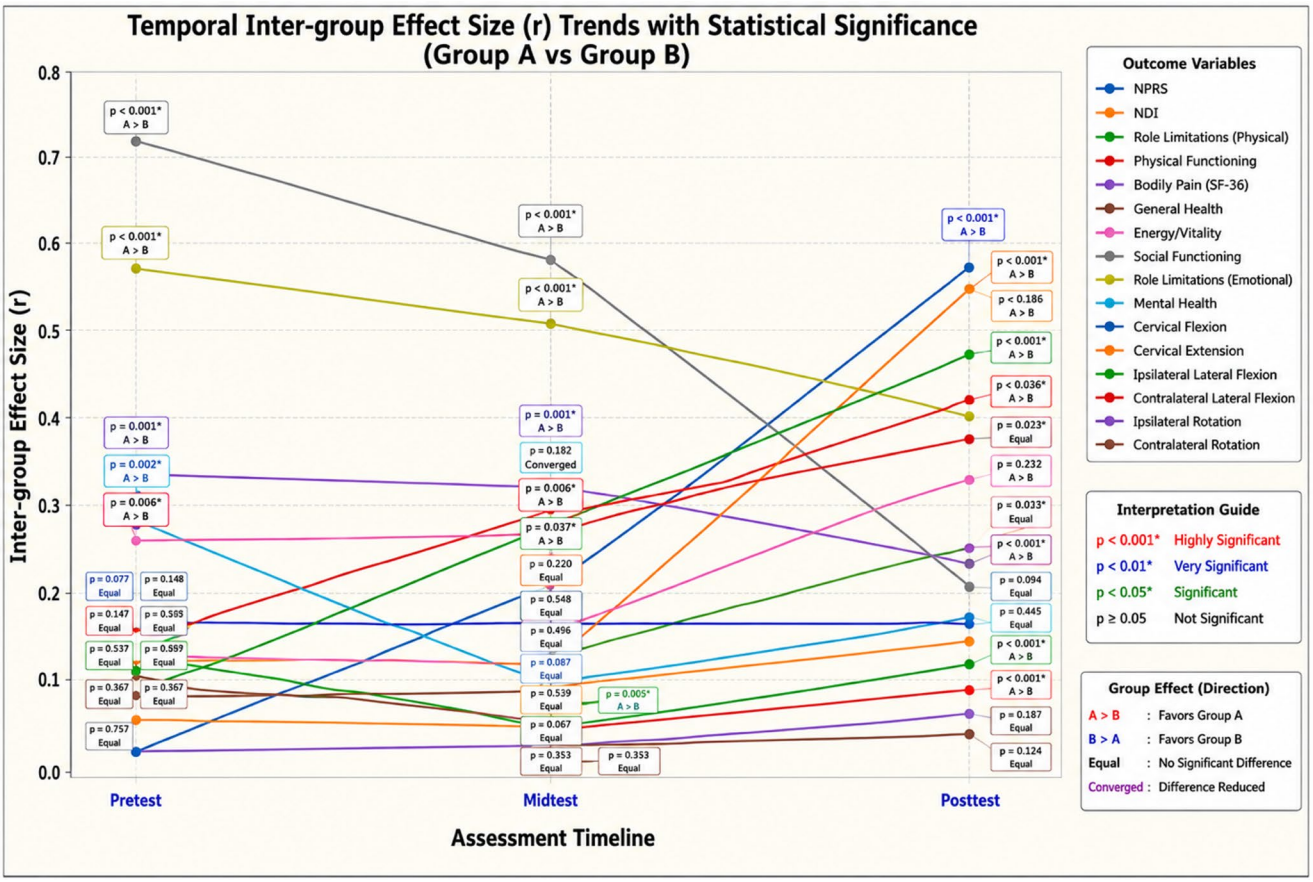


Fig. 3 The line graph illustrates the progression of inter-group effect sizes (r) for pain intensity, functional disability, health-related quality of life (SF-36 domains), and cervical range of motion variables over the study timeline. Effect size values were derived from the Mann–Whitney U test to determine the magnitude of differences between the two intervention groups at each assessment point. Higher r-values with significant p-values indicate stronger inter-group differences (Larger effects of Group A than Group B). Variables demonstrating progressive increases in effect size toward the posttest phase suggest a growing temporal distinction of one intervention over the other. Non-significant values indicate comparable outcomes between the groups

an underestimate of the true treatment effect than an overestimate. Moreover, significant inter-group differences and moderate-to-large effect sizes suggest that the principal study conclusions remained robust despite the imputation approach.

Both treatment groups demonstrated statistically significant reductions in ***NPRS scores*** across all three assessment timepoints, with within-group Pre→Post effect sizes classified as very large in both groups. Within Group B, the Pre→Mid (Z) score for pain intensity was the largest early-phase within-group effect gain recorded across either group or any outcome, and the Friedman χ^2 value for NPRS was the highest in the entire Group B dataset, findings that collectively confirm that DN + RPT produced the most statistically concentrated and powerful within-group early pain response in the study. This observation is consistent with systematic review and meta-analysis evidence reporting that dry needling provides significant short-term reductions in pain intensity associated with myofascial trigger points [7, 15]. Previous randomized controlled trials have similarly demonstrated that dry needling applied to upper trapezius trigger points can significantly reduce pain intensity in patients with cervical myofascial pain [16]. Systematic reviews and meta-analyses have consistently reported that dry needling is associated with significant short-term pain reduction in myofascial pain disorders, though the magnitude of these effects has been characterised as moderate and heterogeneous across studies due to variations in needling protocol, trigger point selection, and concomitant intervention [15, 17]. *Between-group comparisons* revealed a clearer and more clinically meaningful understanding of treatment response than the within-group results alone. At pretest, the groups were statistically equal on NPRS, confirming comparable baseline pain levels and attributing all subsequent divergence to the differential effects of the interventions. A significant between-group difference emerged by midtest in favour of Group A, despite Group B's numerically larger within-group early (Z) score, an apparent contradiction that

becomes clear on understanding that temporal change within the group and endpoint positioning between the groups are different statistical entities altogether. The between-group effect at post-test reached a large effect size, surpassing the NPRS MCID threshold of ≥ 2 points [18], for neck pain patients, thus constituting a clinically meaningful benefit of LLLA that could not be accounted for by chance alone. Clinical trials of laser therapy for chronic cervical pain report moderate-quality evidence for pain relief that is most pronounced at intermediate and final assessments, particularly with appropriate photobiomodulation dosing parameters [13], and meta-analytic evidence has similarly demonstrated beneficial effects of low-level laser therapy on neck pain outcomes in musculoskeletal conditions [19]. Recent systematic reviews on Low-Level Laser Acupuncture have further documented improvements in musculoskeletal pain and trigger point sensitivity through comparable photobiomodulatory pathways [12].

NDI scores improved substantially in both groups across all three assessment timepoints, with Pre→Post within-group effect sizes very large in both groups, and Pre→Post (Z) values among the largest recorded across any outcome in either group. Group B produced a large Pre→Mid (Z) score for disability, confirming that DN + RPT generated the most rapid and statistically powerful early disability reduction in the study, a finding consistent with prior evidence that Dry Needling can rapidly reduce disability and improve neuromuscular function by decreasing trigger point irritability and restoring normal muscle activation patterns [7, 15]. Critically, Group B's Mid→Post (Z) for NDI exceeded Group A's, and its Friedman χ^2 value exceeded Group A's by 1.7 points, confirming that DN + RPT disability gains were not only rapid but genuinely sustained and distributed more evenly across both treatment phases. *Between-group comparisons*, however, revealed a pattern of late divergence structurally distinct from the pain trajectory: no significant intergroup difference was detected at pretest or midtest, indicating statistical equivalence of disability reduction through the mid-treatment phase, before a large between-group effect emerged exclusively at posttest, surpassing the NDI MCID of 7–10 points [7]. This late-surge pattern, equivalence through mid-treatment followed by substantial posttest divergence in favour of LLLA + RPT, indicates that LLLA's disability benefits consolidate progressively in the latter phase of treatment and would be missed entirely by shorter intervention programmes or designs omitting the posttest measurement. Comparable late-emerging functional advantages have been reported in randomised trials of photobiomodulation in musculoskeletal conditions, where the cumulative tissue-level effects of repeated laser application, including collagen remodelling, neuromuscular re-education, and progressive cellular repair, require a sufficient therapeutic dose before translating into perceptible functional gains [13, 19]. A randomised trial by Öz N et al. reported that low-level laser therapy produced significantly greater functional improvements than a sham intervention at final assessment but not at mid-treatment, directly paralleling the late-divergence pattern observed in the present NDI data [13]. The present findings therefore support the view that both DN and LLLA contribute meaningfully to functional improvement within multimodal rehabilitation, but that their contributions differ temporally in clinically important ways: DN accelerates early disability reduction, while LLLA drives the sustained functional consolidation responsible for the larger endpoint advantage.

Both groups demonstrated statistically significant and exceptionally large within-group improvements across all six ***cervical ROM*** directions at every assessment timepoint. Group B produced consistently higher Friedman χ^2 values than Group A and larger Pre→Mid (Z) scores across all ROM directions, confirming that DN + RPT generated stronger and more temporally concentrated early within-group mobility gains, an observation mechanistically coherent with its capacity for immediate myofascial trigger point deactivation and rapid relief of muscle contracture and fascial shortening [9, 20]. Despite this early advantage, *between-group comparisons* revealed no significant intergroup differences for cervical flexion, extension, or either rotation at any timepoint, suggesting that when both techniques are delivered alongside RPT, which itself promotes cervical mobility through therapeutic exercise and postural correction, their differential effects on axial cervical mobility converge toward a common functional endpoint, with laser-associated gains through reduced hypertonicity and improved local circulation [13, 19] ultimately producing equivalent outcomes to the neurophysiological deactivation mechanism of DN. Ipsilateral lateral flexion was the notable exception, demonstrating progressive intergroup divergence, no baseline difference, a significant midtest advantage for Group A, and a near-large posttest effect, alongside the largest within-group Δr between groups across all ROM outcomes (Group A $r = 0.927$ versus Group B $r = 0.847$, $\Delta r = +0.080$), consistent with LLLA's cumulative photobiomodulatory effects on lateral cervical soft-tissue extensibility and collagen remodelling. Contralateral lateral flexion carried a significant pretest difference ($r = 0.274$, $p = 0.006$, 95% CI), indicating baseline imbalance, and its posttest advantage ($r = 0.403$, $p < 0.001$, 95% CI) should therefore be interpreted with caution.

Across all eight ***SF-36 domains***, both groups demonstrated statistically significant and exceptionally large *within-group* improvements, confirming the clinical meaningfulness of both interventions for patient-reported well-being. Within Group A, Friedman χ^2 values followed a

descending physical-to-psychosocial gradient, from SF-36 Bodily Pain to Mental Health, reflecting LLLA's proportionally stronger temporal differentiation in physical symptom domains, consistent with photobiomodulation's primary peripheral tissue mechanism. Group B replicated this gradient with uniformly higher χ^2 values and larger Pre→Mid (Z) scores across all domains, confirming Dry Needling's stronger early within-group quality-of-life response, particularly on SF-36 Bodily Pain (Pre→Mid Z = − 6.73), consistent with the previous studies reporting the rapid somatic symptom relief characteristic of trigger point neurophysiology [9, 16]. *Between-group comparisons* revealed domain-specific trajectories: Physical Functioning and Role Limitations (Physical) diverged progressively in favour of LLLA + RPT, reaching medium posttest effects, alongside a small Vitality advantage, confirming that LLLA's physical quality-of-life benefits are dose-dependent and accumulate toward endpoint [13]. Role Limitations (Emotional) demonstrated the most durable between-group advantage in the dataset, with large effects favouring LLLA + RPT sustained across all three timepoints, possibly reflecting systemic neuromodulatory influences of photobiomodulation on affective regulation in chronic pain. Social Functioning and Mental Health showed initial LLLA + RPT advantages that resolved by posttest, converging to non-significance, suggesting dry needling achieves equivalent psychosocial outcomes on these domains through progressive central sensitisation normalization [4]. It should be noted that significant pretest imbalances existed for Social Functioning, Role Limitations (Emotional), Mental Health, and SF-36 Bodily Pain. *Posttest between-group findings* for these domains require cautious interpretation, and future trials should stratify randomisation accordingly. General Health showed no significant between-group difference at any timepoint. Overall, both interventions produced broadly equivalent SF-36 outcomes by the end of treatment, with LLLA + RPT demonstrating sustained improvements in physical functioning domains and Role Limitations (Emotional), a pattern consistent with the broader literature [13].

Dry needling exerts its rapid analgesic effects through several convergent neurophysiological mechanisms: mechanical disruption of the dysfunctional contracture knot at the motor endplate, suppression of spontaneous electrical activity and reduction in acetylcholine concentration, local release of endogenous opioids, bradykinin, serotonin, and substance P, and activation of A-delta and C-fibre afferents that engage supraspinal descending inhibitory pain modulation [4, 9]. The local twitch response elicited during needling is associated with an immediate reduction in nociceptive sensitisation at the trigger point and a measurable acute increase in pressure pain threshold, effects that manifest within the session and accumulate rapidly across the first several treatments [4]. *Low-level laser Acupuncture* operates through a fundamentally different biological timeline. Photobiomodulation enhances mitochondrial activity through cytochrome c oxidase photoactivation, increases adenosine triphosphate production, improves microcirculation and local tissue oxygenation, modulates reactive oxygen species, and initiates anti-inflammatory cascades through suppression of prostaglandin E2, interleukin-1β, and tumour necrosis factor-alpha [11–19]. These cellular and biochemical processes are cumulative and dose-dependent: each treatment session adds incremental photonic energy to the tissue, with therapeutic effects building progressively across sessions rather than manifesting acutely after a single intervention. The additional stimulation of acupuncture points with laser may further enhance treatment efficacy through activation of endogenous opioid systems, central pain modulation, and descending inhibitory pathway recruitment [3, 4], while the photobiomodulatory effects on connective tissue, including collagen remodelling, fibroblast activation, and progressive soft-tissue extensibility, provide a structural substrate for the selective ipsilateral lateral flexion ROM advantage observed in LLLA at posttest [19].

Similar effects have been found in previous studies on both DN and laser-based interventions [3, 4, 11, 19, 20]. Conversely, in the present study, differences in the rate and pattern of improvement were found in some of the outcome measures between the interventions. These differences could be due to variations in the interventions' protocols, dosage parameters, and the addition of acupuncture point stimulation alongside conventional physical therapy. Additionally, individual variability in trigger point sensitivity and neuromuscular response may influence the magnitude and timing of clinical improvement. However, the lack of a sham/placebo group and the short follow-up period mean that the findings should be taken as indicative of comparative clinical response rather than as evidence of long-term treatment superiority.

Another crucial factor is the implementation of LLLA and DN as part of a multimodal rehabilitation regimen. According to current evidence-based guidelines for treating chronic neck pain, there is a need to use an integrated combination of targeted therapy with therapeutic exercise and manual techniques, as well as patient education, in order to tackle complex causes of myofascial pain syndromes [1]. The inclusion of physical therapy, which consisted of exercise-based rehabilitation, in the care program for both groups was probably the reason for the effectiveness of the treatment, leading to the positive outcomes for cervical joint movement and functions. Similar findings have been reported in previous trials demonstrating that the combination of trigger point therapy with active approaches can

yield even better results compared to passive therapy alone [16, 17, 21]. Therefore, the clinical benefits observed in this study may reflect the synergistic effects of localized trigger point treatment, neuromodulatory stimulation of acupuncture points, and active physical rehabilitation.

An important methodological feature of the present study was the application of both DN and LLLA not only on myofascial trigger points but also on selected acupuncture points in order to integrate: local mechanical dysfunction correction (via MTrPs) and segmental and systemic neuromodulation (via acupuncture points). This combined approach is also consistent with several clinical trials in myofascial pain syndromes where both trigger point needling and acupuncture point stimulation were used within the same treatment framework, reporting meaningful improvements in pain and function [3, 4].

While the current results indicate positive impacts of both treatments on alleviating pain, functional disability, range of motion, and quality of life, the existing literature presents some variability in the magnitude and consistency of treatment effects. For example, systematic reviews and meta-analyses found that Dry Needling is associated with short-term relief from myofascial pain and enhances pressure pain threshold and disability; nevertheless, the magnitude of these effects was considered moderate and heterogeneous across studies [15, 17]. Likewise, randomized clinical trials documented that dry needling was capable of eliciting clinically relevant improvements in pain and cervical functions, but other studies revealed similar therapeutic impacts to other conservative procedures like manual therapy or exercise programs [8, 21]. In addition, conflicting results have been found regarding Low-Level Laser Therapy, specifically when used on acupuncture points and trigger points. In some cases, there may be variations in the effectiveness of the procedure due to differences in the parameters used in the treatment, such as wavelength, dosage, application site, and time [12, 16, 22]. Variability among patients and the methodology used in trials can be considered other potential reasons for inconsistencies. It seems like, although both approaches proved effective, it is quite possible that their comparative effectiveness depends on certain conditions, dosage of treatment, and complementary approaches used together.

Taking into account the results of this investigation as a whole, it is reasonable to conclude that combining Dry Needling and Low-Level Laser Acupuncture may be effective within a complex rehabilitation strategy employed in cases of chronic cervical myofascial pain syndrome. Thus, regarding the speed of recovery in terms of pain and disability, the combination of DN + RPT was more efficient compared to LLLA + RPT. At the same time, the combination of LLLA + RPT produced gradual changes in several indicators. In this connection, it should be noted that the findings obtained in this investigation correlate with the literature data, according to which the effectiveness of Dry Needling can be explained by rapid pain relief based on neurophysiological principles, while LLLA can produce cumulative effects because of biological processes [9, 14, 19, 20]. From the practical perspective, the selection of a particular combination of interventions depends on the objectives pursued in a specific period of time, such as immediate pain reduction versus sustained functional and psychosocial improvements.

Furthermore, the use of both trigger points and acupuncture points in this therapy represents a very important clinical application, where modern myofascial pain therapies have been successfully integrated with more traditional methods of neuromodulation. Hence, the combined treatment approach may account for the wide array of benefits observed concerning pain management, functional and mobility improvement, as well as quality of life enhancement. This integrative perspective aligns with the growing body of literature advocating for multidimensional treatment strategies that address both peripheral myofascial dysfunction and central pain modulation in chronic musculoskeletal conditions.

The current investigation adds to the body of clinical research that has explored the relative use of DN and LLLA along with RPT in the treatment of CCMPS. Rather than suggesting that one treatment technique is more effective than the other, the current results suggest that both DN and LLLA may have benefits as part of a multimodal physical therapy program, with DN producing earlier benefits in terms of patient symptoms and LLLA producing continued benefits in certain patient-reported outcomes.

Theoretical and clinical consequences

Based on the research results, it can be suggested that CMPS should be treated through a multifactorial and time-dependent intervention strategy. Dry needling, combined with RPT, is the intervention of choice when the clinical priority is rapid early reduction in pain intensity, functional disability, and cervical mobility, while LLLA, combined with RPT, is the intervention of choice when the clinical priority is sustained, comprehensive, and progressively accumulating improvement across pain, disability, physical quality of life, emotional well-being, and lateral cervical mobility. Moreover, it is a safe and integrative technique that can serve as a rehabilitation technique in the long run. The smaller gap between the measurements of pain and anxiety levels after LLLA administration additionally supports its role in the neuroimmune and CNS modulation mechanisms.

These distinctions increase the developing comprehension of multimodal frameworks in the management of chronic myofascial pain.

The study's strengths

The study has numerous strengths that make it clinically significant. First, this research compared two commonly used trigger point therapy techniques, which provided insights into their respective clinical responses. Second, the use of different measurements, including pain, disability, neck range of motion, and HRQoL, provided a detailed assessment of the effectiveness of the intervention. Third, the use of both trigger point therapy and acupuncture points can be seen as an application approach for myofascial pain management. Lastly, the appropriate use of non-parametric statistics enhanced the reliability of the results due to the nature of the data distribution.

The study's limitations

Some limitations need to be considered in this study. (1) The small sample size, non-probability convenience sampling, use of LOCF, and lack of multicenter design make the results less generalizable. (2) A formal complete-case sensitivity analysis restricted to the 91 participants with full data was not conducted; however, given the minimal dropout volume, the non-outcome-related nature of withdrawals (MCAR), and the conservative directional bias of LOCF in a progressive improvement context, the primary conclusions are considered robust to the method of missing data handling. (3) The lack of a placebo/sham group and a single-assessor blind design raises the risk of potential expectancy. (4) Besides, the technique's effectiveness is associated with practitioners' skills, so a slight difference in its implementation could affect outcomes, raising the risk of performance bias. (5) Use of subjective self-reported measures, lack of pressure algometry test or other objective measures of pain sensitivity, and the short follow-up period. (6) Additionally, despite randomisation, statistically significant pretest differences were identified for five outcome variables, SF-36 (social functioning, role limitations (Emotional), Mental Health, Bodily Pain), and Contralateral Lateral Flexion, all of which favoured Group A (LLLA+RPT) at baseline. Posttest between-group advantages observed on these specific outcomes should therefore be interpreted with caution, as the observed differences may partially reflect pre-existing group imbalances rather than treatment effects alone. Finally, individual response to needle insertion, as well as other uncontrolled variables, including psychological stress, work-related posture, and activities, could lead to variance in the treatment efficacy.

Recommendations

It is suggested that there should be more multicenter long-term follow-up RCT studies on a large sample size to ensure better generalization, sustainability of the treatment effects, and to apply more advanced missing-data techniques, such as multiple imputation or mixed-effects modeling, to further enhance statistical precision, sensitivity, and methodological rigor. Objective measurements like inflammatory biomarkers, electromyography, or neuroimaging techniques would be useful to evaluate the mechanism of pain. Randomized control trials with sham or placebo control groups would be helpful to determine the relative effectiveness of the procedures used in the management of chronic myofascial pain syndrome. Future trials should employ stratified randomisation or baseline-adjusted covariate analyses (e.g., ANCOVA-based sensitivity analysis) on variables with baseline imbalances in order to eliminate the randomization limitation.

Conclusion

The current study examined the effectiveness of two interventions, Low-Level Laser Acupuncture (LLLA) and Dry Needling (DN), alongside routine physical therapy for chronic cervical myofascial pain syndrome. Both techniques resulted in notable changes in pain intensity, functional disability, range of motion, and various dimensions of health-related quality of life throughout the trial. While DN+RPT yielded slightly faster initial results for pain and functional impairment in within-group analysis, LLLA+RPT had more sustained and clinically meaningful results over time in between-group analyses concerning specific patient-reported outcomes, especially those about vitality and psychosocial parameters. It was concluded that the techniques could be used as effective additions to routine physical therapy in a comprehensive rehabilitation program. However, due to the absence of a sham or placebo-controlled group and the relatively short follow-up period, the results should be interpreted as evidence of comparative clinical effectiveness rather than definitive superiority of one technique over the other.

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Data availability All data generated or analyzed during this study are included in this article.

Declarations

Name of the approval committee (IRB) Ethical approval for the study was obtained from the Institutional Review Board/Ethics Committee of the University of Lahore, ensuring compliance with established ethical principles and participant protection. (REC No. 443, approved on 1ST December 2022).

Human ethics This randomized controlled trial was conducted in accordance with the ethical principles of the *Declaration of Helsinki for research involving human participants*.

Consent to participate All participants signed a written informed consent form before inclusion in the study.

Clinical trial registration The clinical trial was prospectively registered in the Iranian Registry of Clinical Trials (IRCT) under the Registration No. (IRCT20190618043930N2). The registry record is publicly accessible through the IRCT database.

Trial Registration Iranian Registry of Clinical Trials (IRCT), Registration No. IRCT20190618043930N2.

Competing interests The authors declare no competing interests.

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